Geometric areas

1

a 12 units^2 b 21 units^2 c 4 units^2 d 12 units^2

2

a 18 b 72 c 104 d 38.5

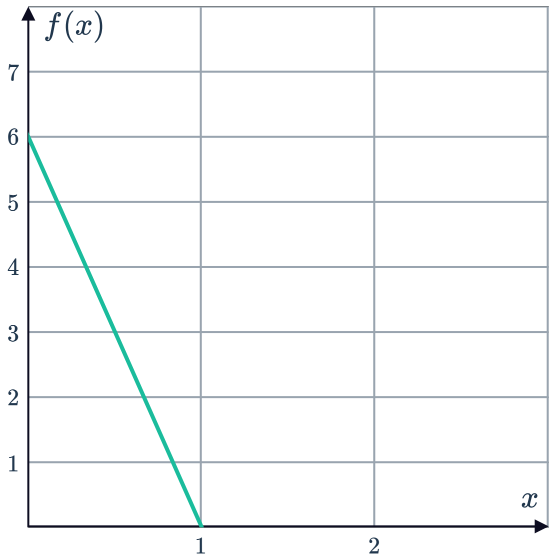
e 18 f 80 g 18π h 9π

i 32π j 18π

3

a

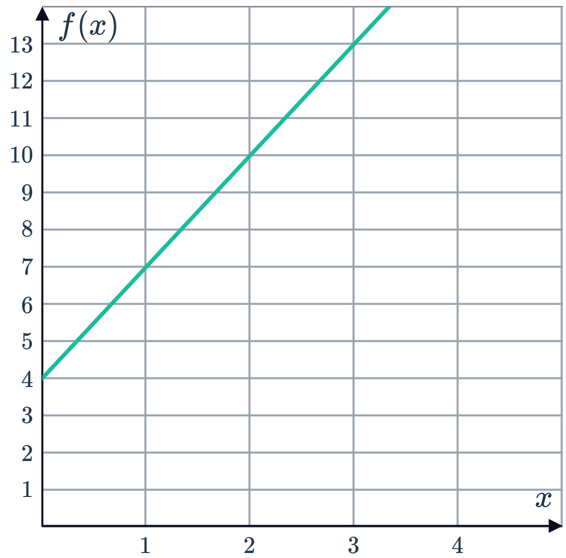
i



ii 3 units^2

b

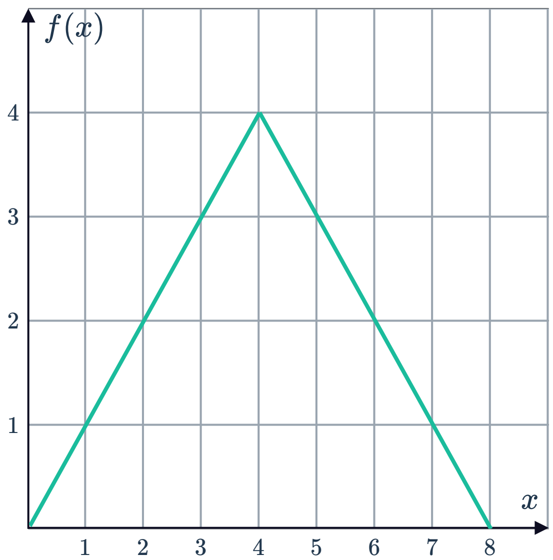
i



ii $\frac{51}{2} \text{ units}^2$

c

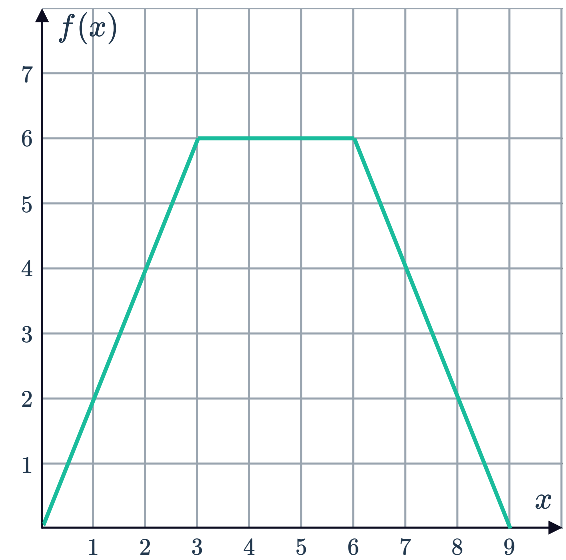
i



ii 16 units^2

d

i



ii 36 units^2

Approximate areas

4

a An overestimation as the function is increasing on the given interval.

b An underestimation as the function is decreasing on the given interval.

5

a An underestimation as the function is increasing on the given interval.

b An overestimation as the function is decreasing on the given interval.

6 The left endpoint approximation because the function is always increasing.

7

a

i Overestimate

ii Underestimate

iii Underestimate

iv Overestimate

b

i Underestimate

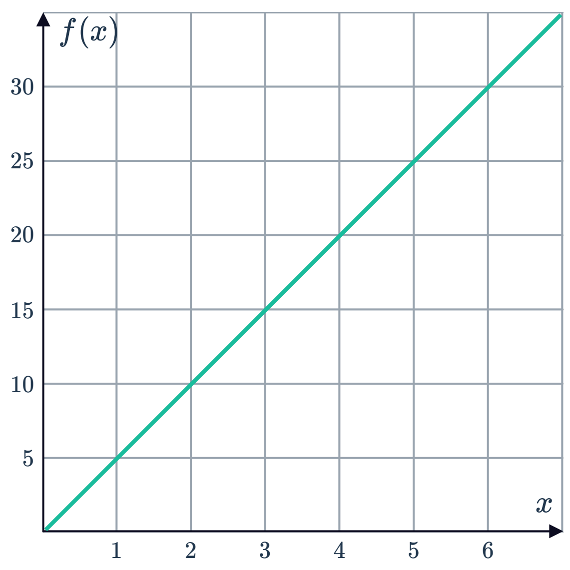
ii Overestimate

iii Overestimate

iv Underestimate

8

a



b

i 60 units^2

ii 120 units^2

c

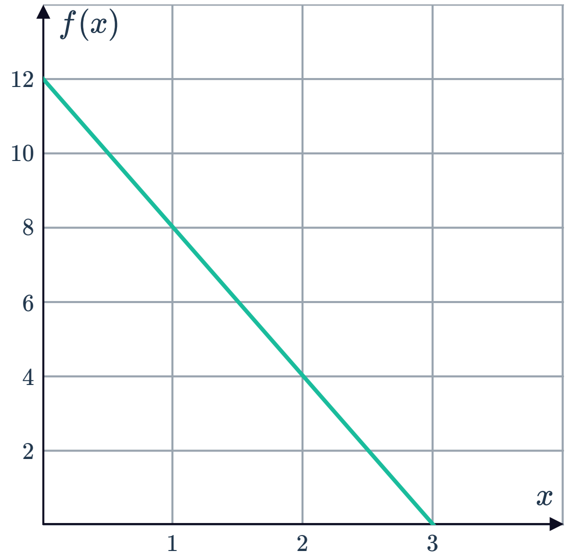
i 75 units^2

ii 105 units^2

d $A = 90 \text{ units}^2$

9

a



b **i** 24 units^2 **ii** 12 units^2

c **i** 21 units^2 **ii** 15 units^2

d 18 units^2

10

a 62 units^2 **b** 50 units^2

11 336 units^2

12 $\frac{77}{60}$ units^2

13

a 432 units^2 **b** 576 units^2

Use technology

14

a 30 **b** $e^3 - 1$ **c** 36

15

a

n	5	10	100	1000	10 000
A_L units^2	3.560	3.940	4.293	4.329	4.333

b $\frac{13}{3}$ units^2

16

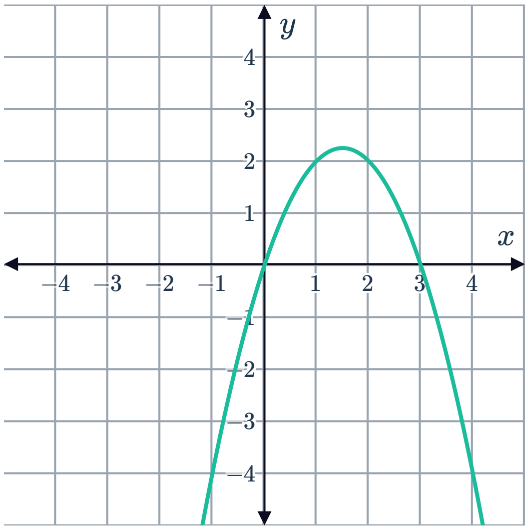
a

n	5	10	100	1000
A_R units^2	2.63705	2.90452	3.12042	3.13956

b π units^2

17

a



b $0 \leq x \leq 3$

c

n	5	10	100	1000	10 000
A_R units^2	4.3200	4.4550	4.4996	4.5000	4.5000

d $\frac{9}{2}$ units^2